

# KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

## MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

**Subject: Database Management Systems (DBMS)**

**Time: 1.5 Hours**

**Semester / Programme: 4th Semester**

**Full Marks: 20**

**Instructions:** Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

### SECTION - A (Compulsory)

- 1(a)** What is data independence in DBMS? Differentiate between logical and physical data independence. [1]
- 1(b)** State the cardinality ratio and participation constraint for a binary relationship set 'Enroll' between student entity set and course entity set. [1]
- 1(c)** Define super key and candidate key. Is every candidate key a super key? [1]
- 1(d)** What is the role of the  $\sigma$  (sigma) and  $\pi$  (pi) operators in Relational Algebra? [1]  
 $\sigma_{condition}(R)$  and  $\pi_{attributes}(R)$
- 1(e)** Given a relation schema  $R(A, B, C, D)$  with functional dependency set  $F = \{A \rightarrow B, B \rightarrow C\}$ , compute  $(A^+)$  [1]  
 $A^+$

### SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

- Q.2. (a)** Explain the 3-schema architecture of a Database Management System. What is the primary objective of this architecture? [2.5]
- (b)** Discuss the concepts of Specialization and Generalization in E-R modeling with suitable examples and appropriate diagrammatic notations. [2.5]
- Q.3. (a)** Consider the following relational schema for a University Database: Student (RollNo, Name, Dept, Semester) Course (CourseID, CourseName, Credits) Enroll (RollNo, CourseID, Marks) Write SQL queries for the following: (i) Find the names of all students enrolled in the course named 'Database Systems'. (ii) Find the total number of credits for courses taken by the student with RollNo 'KIIT101'. [3]
- (b)** Explain entity integrity and referential integrity constraints with examples in relational databases. [2]
- Q.4. (a)** State and prove the Pseudo-transitivity rule using Armstrong's Axioms on functional dependencies. [3]  
Formula:  $X \rightarrow Y$  and  $Y \rightarrow Z$  implies  $X \rightarrow Z$
- (b)** Given two sets of functional dependencies  $F_1 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$  and  $F_2 = \{A \rightarrow B, B \rightarrow C\}$ , check whether  $(F_1 \equiv F_2)$  (are they equivalent?). [2]  
 $F_1^+ = F_2^+$

- Q.5.** (a) What is normalization? Explain the need for normalization and define First Normal Form (1NF) with a suitable counter-example. [2.5]
- (b) Explain Second Normal Form (2NF) and Third Normal Form (3NF) with formal definitions. [2.5]  
 $X \rightarrow A$
- 

--- END OF PAPER ---